

Mode

Definition

The **Mode** is a measure of central tendency that identifies the **most frequently occurring value** in a dataset. Unlike the mean (average) or median (middle value), the mode is simply the value that appears the highest number of times.

A dataset can have:

- **One Mode (Unimodal):** If one value appears most frequently. (e.g., 1, 2, 2, 3, 4)
- **Two Modes (Bimodal):** If two different values have the same highest frequency. (e.g., 1, 2, 2, 3, 3, 4)
- **Multiple Modes (Multimodal):** If more than two values share the same highest frequency.
- **No Mode:** If all values occur with the same frequency (e.g., 1, 2, 3, 4, 5).



Formula

There is **no single calculation formula** for the mode in the same way there is for the mean.

- For **discrete data**, the mode is found by **inspection** (counting frequencies).
- For **continuous frequency distribution**, a specific formula is used to *estimate* the mode from the modal class.

Continuous Frequency Distribution Formula:

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

(The components of this formula are explained in the continuous distribution section below).



Examples

In Case Of Discrete Distribution

This category includes both raw, ungrouped data and data organized into a discrete frequency table.

Case 1: Ungrouped Data (Raw List)

Procedure: Count the occurrences of each number and find the one that appears most.

- **Example (Unimodal):**
 - **Data:** 5, 7, 8, 2, 7, 10, 7
 - **Frequencies:** 5 (1), 8 (1), 2 (1), 10 (1), 7 (3 times)
 - **Mode = 7**
- **Example (Bimodal):**
 - **Data:** 10, 15, 12, 10, 18, 15, 11
 - **Frequencies:** 10 (2 times), 15 (2 times), 12 (1), 18 (1), 11 (1)
 - **Mode = 10 and 15**
- **Example (No Mode):**
 - **Data:** 1, 2, 3, 4, 5, 6
 - **Frequencies:** All values appear only once.
 - **Mode = No Mode**

Case 2: Discrete Frequency Distribution (Table)

Procedure: Look for the **highest frequency** (f) in the table. The **mode** is the corresponding value (x) in that row.

Example:

x (Value)	f (Frequency)	
10	2	
20	5	
30	8	← Highest Frequency
40	3	
50	1	

1. **Find Highest Frequency:** The highest frequency (f) is **8**.
2. **Find Corresponding x :** The x value for $f = 8$ is **30**.
3. **Mode = 30**



In Case Of Continuous Frequency Distribution

In this case, data is grouped into class intervals (e.g., 10 – 20). We cannot find an exact mode, so we **estimate** it using a formula.

Procedure:

1. **Identify the Modal Class:** This is the class interval with the **highest frequency** (f_1).
2. **Apply the Mode Formula:**

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

Explanation of Formula Components:

- L = **Lower limit** of the Modal Class.
- f_1 = **Frequency** of the Modal Class (the highest frequency).
- f_0 = **Frequency** of the class **preceding** (before) the Modal Class.
- f_2 = **Frequency** of the class **succeeding** (after) the Modal Class.
- i = **Class width** (Upper limit - Lower limit) of the Modal Class.

Example:

Class Interval	f (Frequency)	
10 – 20	5	(f_0)
20 – 30	15	$(f_1) \leftarrow$ Modal Class
30 – 40	8	(f_2)
40 – 50	4	

1. **Identify Modal Class:** The highest frequency is **15**, so the Modal Class is 20 – 30.

2. **Identify Components:**

- $L = 20$ (Lower limit of 20 – 30)
- $f_1 = 15$ (Frequency of 20 – 30)
- $f_0 = 5$ (Frequency of the class *before*, 10 – 20)
- $f_2 = 8$ (Frequency of the class *after*, 30 – 40)
- $i = 10$ (30 – 20)

3. **Apply Formula:**

$$\text{Mode} = 20 + \frac{15 - 5}{2(15) - 5 - 8} \times 10$$

$$\text{Mode} = 20 + \frac{10}{30 - 13} \times 10$$

$$\text{Mode} = 20 + \frac{10}{17} \times 10$$

$$\text{Mode} = 20 + \frac{100}{17}$$

$$\text{Mode} = 20 + 5.88$$

$$\text{Mode} \approx \mathbf{25.88}$$

The estimated mode is 25.88, which falls logically within the modal class (20 – 30).

Advanced Note: Empirical Relationship

For distributions that are moderately skewed (not symmetrical), there is an approximate relationship between the mean, median, and mode:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$$

This is often used to estimate the mode if the mean and median are already known.

This method is primarily used for **grouped frequency data** when the highest frequency is not immediately clear or there are two high, close frequencies (bi-modal tendency).

Example: Confirming the Modal Class (Analysis Table)

Problem: Find the mode for the following data distribution.

Class Interval	Frequency (f)
10-20	5
20-30	12
30-40	14
40-50	13
50-60	9

Step 1: Create the Analysis/Grouping Table

We create a table with six columns based on how the frequencies are grouped:

- **Col 1:** Original Frequency (f)
- **Col 2:** Grouping by 2s (sum of $f_1 + f_2, f_3 + f_4, \dots$)
- **Col 3:** Grouping by 2s starting with the 2nd frequency (sum of $f_2 + f_3, f_4 + f_5, \dots$)
- **Col 4:** Grouping by 3s (sum of $f_1 + f_2 + f_3, f_4 + f_5 + f_6, \dots$)
- **Col 5:** Grouping by 3s starting with the 2nd frequency (sum of $f_2 + f_3 + f_4, \dots$)
- **Col 6:** Grouping by 3s starting with the 3rd frequency (sum of $f_3 + f_4 + f_5, \dots$)

Class Interval	Col 1 (f)	Col 2	Col 3	Col 4	Col 5	Col 6
10-20	5	17 (5+12)		31 (5+12+14)		
20-30	12		26 (12+14)		39 (12+14+13)	

Class Interval	Col 1 (f)	Col 2	Col 3	Col 4	Col 5	Col 6
30-40	14	27 (14+13)				36 (14+13+9)
40-50	13		22 (13+9)			
50-60	9					

Step 2: Create the Mode Confirmation Table

In each column (1 to 6), circle the **highest sum** and note which Class Interval(s) contributed to that sum.

Class Interval	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Total Marks
10-20				1			1
20-30		1	1	1	1		4
30-40	1	1	1	1	1	1	6
40-50			1		1	1	3
50-60						1	1

Step 3: Conclusion

The class interval **30-40** received the highest number of tally marks (6). Even though the original frequency (14) was very close to the next highest (13), the grouping method confirms that **30-40** is the **Modal Class**.

Step 4: Calculation

Now we apply the Mode formula using the confirmed modal class:

$$L = 30, f_1 = 14, f_0 = 12, f_2 = 13, h = 10.$$

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

$$\text{Mode} = 30 + \left(\frac{14 - 12}{2(14) - 12 - 13} \right) \times 10$$

$$\text{Mode} = 30 + \left(\frac{2}{28 - 25} \right) \times 10$$

$$\text{Mode} = 30 + \left(\frac{2}{3} \right) \times 10$$

$$\text{Mode} = 30 + \frac{20}{3} \approx 30 + 6.67 = \mathbf{36.67}$$

The mode is **36.67**.